

John Balci

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EDUCATION

Purdue University

West Lafayette, IN

B.S. in Electrical Engineering, Conc. Software Engineering | GPA: 3.58/4.0

Aug. 2025 – May 2028 (Expected)

EXPERIENCE

Software and Instrumentation TRMC Intern

Summer 2026

Naval Test Wing Atlantic, Advanced Systems Group (Hired by ASRC Federal TRMCi)

Patuxent River, MD

- Built and shipped a Python constraint-solving application that replaced a manual **20-40 hour engineering** process with an automated run, now in use by **3+ engineers** at Naval Test Wing Atlantic Instrumentation.
- Implemented a 2D backtracking CSP solver with forward checking, ranking candidate placements with a 6-metric heuristic to pack arbitrary sensor channel sets into IRIG-106 Chapter 4 frames with competing constraints.
- Designed a Tkinter GUI and CLI over the solver with input validation, unit tests, and automated color-coded Excel export; delivered solo and maintained by the group after handoff.

Embedded Software Undergraduate Researcher

Aug. 2025 – Present

Purdue Cubesat Lab, Electric Propulsion and Plasma Lab

West Lafayette, IN

- Authored embedded **C++** firmware for STM32 to translate raw capacitance into real-time volumetric propellant telemetry from custom parallel-plate capacitor application via oversampling and averaging (16–32 samples).
- Synthesized 1V half-wave sinusoidal pulses (20 μ s duration, 1 Hz PRF) in **MATLAB** to characterize discharge cycles for a novel pulsed-magnetoplasma dynamic (PMPD) thruster in a 6U CubeSat.

Founder, Software Engineer – Scrapr.site

Sept 2025 – Present

JEB Digital Ventures

Fairfax, VA

- Architected a **semantic search engine** using OpenAI text embeddings and weighted cosine similarity math to rank faculty profiles against research queries using OpenAlex API publication data **with 26 users**.
- Developed an **end-to-end AI outreach pipeline** transforming candidate CVs and relevant academic metadata into personalized emails and tailored cover letters **based on researcher's recent outputs**.
- Built a cross-platform web and desktop application using Next.js 16, TypeScript, and Electron, integrating Supabase and Clerk for user state management, authentication, and data persistence.

AI Integration Intern

Summer 2025

Chainwise CPA

Reston, VA

- Built 3 scalable **API endpoint integrations** in JavaScript connecting enterprise accounting platforms, automating workflows supporting over \$1.7M in client transactions and saving hours per client deal.
- Designed and implemented 4 real-time AI project management automations using Zapier custom workflows and Claude API, improving operational efficiency.

Lead Propulsion Engineer

2021 – 2025

Caelus Rocketry

Alexandria, VA

- Led a 17-student liquid propulsion team to the complete design, manufacturing, and integration of a liquid-fueled rocket engine, culminating in the successful **completion of cold flow** tests and **full readiness** for static fire.

PROJECTS

Real-Time Asteroid Detection Pipeline | *Python, PyTorch, YOLOv5, Jetson Nano, Git*

2024 – 2025

- Trained and deployed a YOLOv5s computer vision model on edge hardware for real-time moon tracking, optimizing inference for the Jetson Nano's compute and memory budget.

Gyroscopic Stabilized Pen | *C++, OpenCV, ESP32-S3, Raspberry Pi 5, Git*

2024 – 2025

- Built an assistive writing device for users with essential tremor and cerebral palsy, programming a closed-loop computer-vision and microcontroller stabilization system.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, Swift, SQL (Postgres), JavaScript/Typescript, HTML/CSS, MATLAB

Frameworks: React, Node.js, Next.js, Electron, PyTorch, TensorFlow, Numpy, Pandas

Developer Tools: Git, Linux, Clerk, Supabase, REST APIs, OpenAI & Anthropic APIs, STM32/ARM, Arduino

Concepts: Algorithms & Constraint Solving, Semantic Search & Embeddings, Deep Learning, Embedded Systems